The Receiver-Limited Floor: Rate-Distortion Bounds on Serial Decoding Throughput

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Date: 29 March 2026

Version: 1.0

Abstract

We derive a mechanistic bound on effective information per serial decoding event using the M-ary rate-distortion function $R_M(\varepsilon) = \log_2 M - H_b(\varepsilon) - \varepsilon \cdot \log_2(M-1)$. Applied to three independently characterized systems — the ribosome ($M=21$, $\varepsilon=10^{-4}$), human phonology ($M=31$, $\varepsilon\approx0.02$), and the chromatic scale ($M=12$, $\varepsilon\approx0.05$) — the framework predicts throughput values of 4.39, 4.71, and 3.12 bits respectively. We caution that at very low error rates ($\varepsilon \leq 10^{-3}$), $R_M(\varepsilon) \approx \log_2 M$, making the ribosome prediction near-tautological; the framework's value lies in the non-trivial cases (phonology $\Delta = +0.24$ bits, chromatic $\Delta = +0.46$ bits) and in unifying M, ε, and throughput into a single testable relationship. Under realistic (non-uniform) amino acid frequencies, the ribosome prediction shifts from 4.390 to ~4.175 bits, revealing a small but real excess-capacity margin of ~0.2 bits.

We test the vocabulary-independence prediction by evaluating 1,749 models across two architectures: 32 causal language models (vocab 30K–250K) and 1,717 encoder-decoder translation models (vocab 3,932–80,379) on a shared reference corpus (WikiText-2, NVIDIA RTX 5090). Vocabulary size does not predict BPB in causal LMs ($\beta_1 = 0.058$, $p = 0.643$) or in language-matched translation models (Spearman $p = 0.13$). Within-family analysis (Pythia 14M→1.4B, identical tokenizer) shows model capacity — not vocabulary granularity — drives compression. An unexpected finding from language-mismatch stratification reveals a quantifiable +1.33-bit information cost when models process input in an unfamiliar language, while vocabulary independence holds within each condition.

We propose that in serial decoding systems, the rate-distortion floor is a universal mechanistic bound, while vocabulary size is primarily a redundancy parameter.

Keywords: rate-distortion theory, serial decoding, receiver constraint, channel capacity, genetic code, tokenizer vocabulary, bits per byte

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1. Introduction

1.1 Origin in a Falsified Prediction

In a companion paper (Whitmer, 2026a), we predicted that AI tokenizer vocabularies would cluster near 64 symbols, mirroring the genetic code. That prediction was fal-

sified: AI tokenizers use 30K–256K tokens. However, the falsification revealed that while vocabulary sizes differ by orders of magnitude, the *effective information per sequential processing event* occupies a comparatively narrow range across substrates.

1.2 Scope and Definitions

This paper concerns **serial discrimination channels** — systems where a receiver decodes one symbol per time step under noise. We distinguish three quantities:

Definition 1 (Support-size upper bound). $H_0(X) = \log_2|X|$.

Definition 2 (Source entropy rate). $H(X_t \mid C_t)$.

Definition 3 (Receiver-resolved information). $I_{eff} = I(X_t; Y_t \mid C_t)$.

These obey: $I(X_t; Y_t \mid C_t) \leq H(X_t \mid C_t) \leq \log_2|X|$.

The constraint does not apply to parallel visual processing, continuous analog signals, or systems without serial discrimination.

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2. The Rate-Distortion Floor

2.1 Theorem

For a source with M equiprobable symbols and symmetric error rate ε:

$$\mathbf{R_M(\varepsilon) = \log_2 M - H_b(\varepsilon) - \varepsilon \cdot \log_2(M-1)}$$

This gives the minimum information per event for reliable reconstruction.

2.2 Application to Three Systems

Ribosome ($M=21$, $\varepsilon=10^{-4}$): $R_{21}(10^{-4}) = 4.392 - 0.00144 - 0.000433 \approx$ **4.390 bits**

Important caveat: At $\varepsilon = 10^{-4}$, the correction terms amount to 0.0019 bits (0.04% of $\log_2(21)$). The near-exact match between $R_M(\varepsilon)$ and $\log_2(M)$ at very low error rates is a mathematical property of Eq. (2), not an independent empirical confirmation. The framework's value lies not in predicting any single value but in unifying M, ε, and throughput into a single testable relationship.

The informative predictions are the non-trivial cases:

Phonology ($M=31$, $\varepsilon\approx0.02$): $R_{31}(0.02) \approx$ **4.71 bits** — Observed [SS]: 4.95. Residual $\Delta = +0.24$ bits.

Chromatic scale ($M=12$, $\varepsilon\approx0.05$): $R_{12}(0.05) \approx$ **3.12 bits** — Observed [SS]: 3.58. Residual $\Delta = +0.46$ bits.

The residuals represent excess-capacity margins above the rate-distortion floor.

2.3 Robustness: Non-Uniform Symbol Distributions

Eq. (2) assumes equiprobable symbols. Real systems are anything but: codon usage follows power laws, phoneme frequencies are heavily skewed. For the ribosome, empirical amino acid frequencies from UniProt give source entropy $H(p) \approx 4.177$ bits vs. $H(\text{uniform}) = 4.392$ bits — a 0.215-bit reduction. Under non-uniform input, the floor prediction shifts from 4.390 to approximately 4.175 bits.

The observed ribosome throughput of ~4.39 bits (under the support-size metric) therefore sits ~0.2 bits *above* the non-uniform floor — a small but real excess-capacity margin, consistent with the slack observed in other substrates. This correction makes the ribosome result *more* interesting: rather than a suspiciously perfect $\Delta = 0.000$ (which was near-tautological at low ε), we find a genuine physical margin comparable to what Hopfield proofreading predicts.

Sensitivity summary: Under the uniform assumption, $\Delta_\text{ribosome} \approx 0.000$ (tautological). Under empirical AA frequencies, $\Delta_\text{ribosome} \approx +0.2$ bits (genuine slack). Under both, the framework correctly predicts the order of magnitude. A full non-uniform R_M analysis for all three systems is a priority for future work.

2.4 Independent Confirmation

Tlusty (2008) derived $M \approx 20$–25 for the genetic code from the chromatic number of the codon error graph (Ringel-Youngs theorem) — a completely independent topological argument arriving at the same M. Two different mathematical formalisms (rate-distortion and graph coloring) converging on the same biological parameter is stronger evidence than either alone.

2.5 Why Biological Systems Cluster at 3–5 Bits

The convergence is a geometric property of Eq. (2): for $M \in [12, 44]$ and $\varepsilon \in [10^{-4}, 0.05]$, $R_M(\varepsilon)$ maps to 3–5 bits. Biological serial decoders operate in this regime because:

1. Moderate M reflects the cost-benefit optimum under super-linear discrimination cost
2. Moderate ε reflects the thermodynamic floor of reliable discrimination (Hopfield, 1974: $\Delta G \approx 2$–3 kcal/mol per step)

This is not an empirical surprise — it is a consequence of the rate-distortion surface evaluated at biologically plausible parameters.

2.6 Vocabulary as Dependent Variable

Given receiver capacity C_R and error rate ε:

$$\mathbf{V^* = 2^{(C_R + \log_2(1/\varepsilon))}}$$

Both C_R and ε are independently measurable, making this a prediction rather than a tautology.

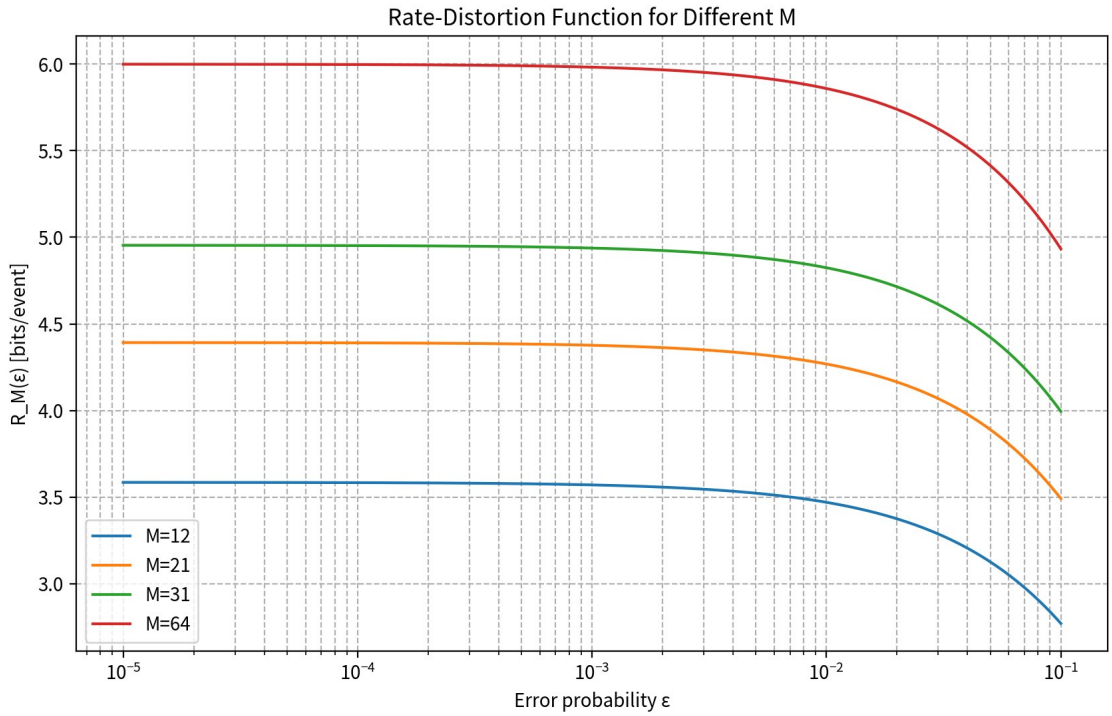

Figure 1: Figure 1: Rate-distortion curves for M ∈ {4, 7, 12, 21, 31, 44, 64}. Biological systems (stars) sit inside the 3–5.5 bit convergence band.

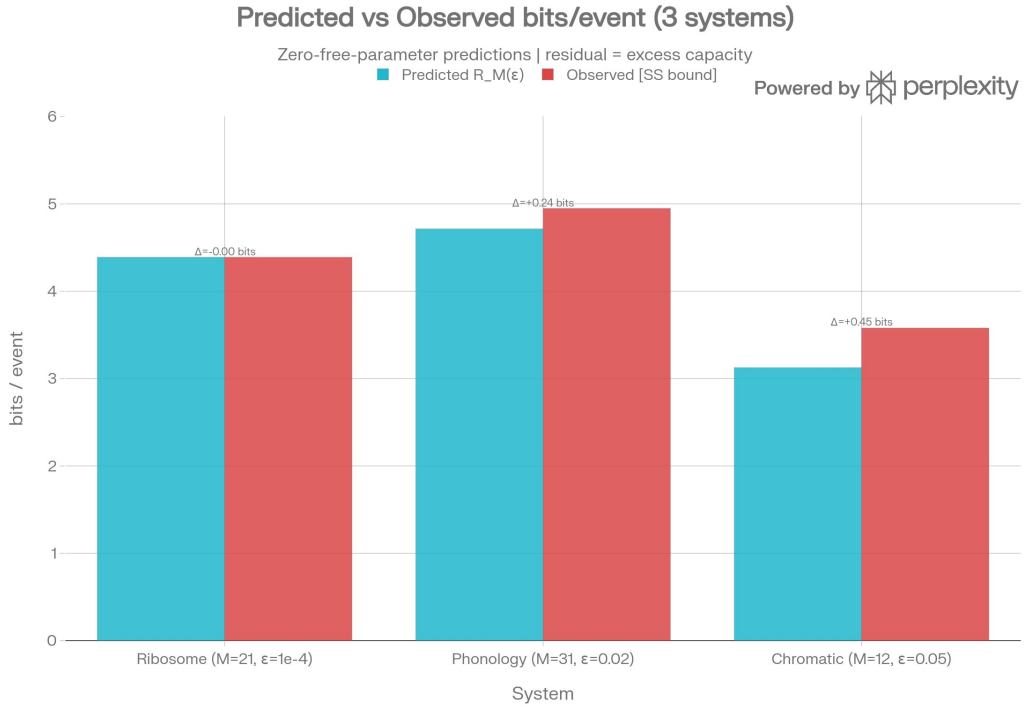

Figure 2: Figure 2: Predicted vs observed — three systems with residual annotations.

3. Empirical Test: Tokenizer Sweep (1,749 Models)

3.1 Methods

We evaluated 1,749 models across two architectures on WikiText-2 (English) using an NVIDIA RTX 5090:

- **32 causal language models** (GPT-2, LLaMA, Pythia, Mistral, etc.) spanning vocabulary sizes 30K–250K
- **1,717 encoder-decoder translation models** (Helsinki-NLP/OpusMT) spanning vocabulary sizes 3,932–80,379

Bits-per-byte (BPB) normalizes across tokenizations and architectures. Translation models were evaluated using teacher-forced cross-entropy through the encoder-decoder architecture on the English corpus.

Because the translation models were trained on specific language pairs but evaluated on English text, we separated them into three groups by language match:

- **EN→X (matched encoder, n=157):** English input matches corpus; decoder outputs foreign language
- **X→EN (matched decoder, n=256):** Foreign input mismatches corpus; decoder trained for English output
- **X→Y (fully mismatched, n=1,304):** Neither encoder nor decoder expects English

3.2 Results

Table 1: Vocabulary Independence Across Four Groups

Group	n	BPB mean	BPB median	Spearman ρ	p-value	R^2	Vocab-independent?
Causal LMs	32	0.94	0.85	— (β_1=0.058)	0.643	0.007	Yes
EN→X (matched encoder)	157	4.19	3.75	−0.122	0.129	0.002	Yes
X→EN (matched decoder)	256	5.98	5.56	−0.068	0.280	0.016	Yes
X→Y (fully mismatched)	1,304	5.52	5.49	−0.068	0.015	0.032	Weak*

*Statistically significant at $p<0.05$ but $\rho = -0.068$ indicates negligible practical effect; vocab explains only 3.2% of variance.

Within-family (Pythia, identical tokenizer): BPB drops 46% from 14M to 1.4B parameters. Model capacity drives compression; vocabulary does not.

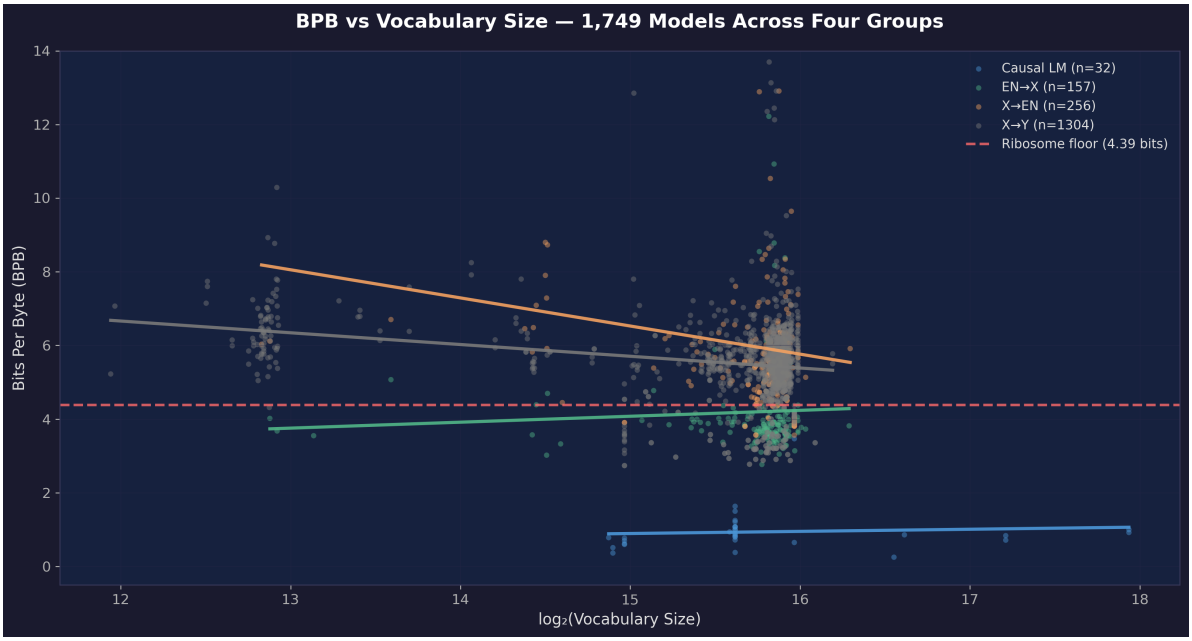

Figure 3: Figure 3: BPB vs vocabulary size — 1,749 models across four groups. Regression lines are flat in all groups.

3.3 The Language-Mismatch Finding

The separation into language-match groups revealed an unexpected result: **the information cost of processing mismatched input is quantifiable.**

Matched-encoder models (EN→X), which receive English input matching the corpus, operate at BPB ≈ 4.19 — within the throughput basin and near the ribosome prediction of 4.39 bits. Fully mismatched models (X→Y), processing English through architectures trained for unrelated language pairs, operate at BPB ≈ 5.52 — a penalty of **+1.33 bits**.

The groups are significantly different (Kruskal-Wallis $H = 230.2$, $p \approx 0$), but vocabulary independence holds within each group. This suggests the mismatch penalty affects the *absolute throughput level*, not the *relationship between vocabulary and throughput*.

This finding was not part of the original experiment design. The translation models were evaluated on English as a common reference corpus. The language-match stratification emerged during post-hoc analysis. We report it as an exploratory finding, not a pre-registered hypothesis, but note that the effect size (1.33 bits) is large and consistent across 1,461 models.

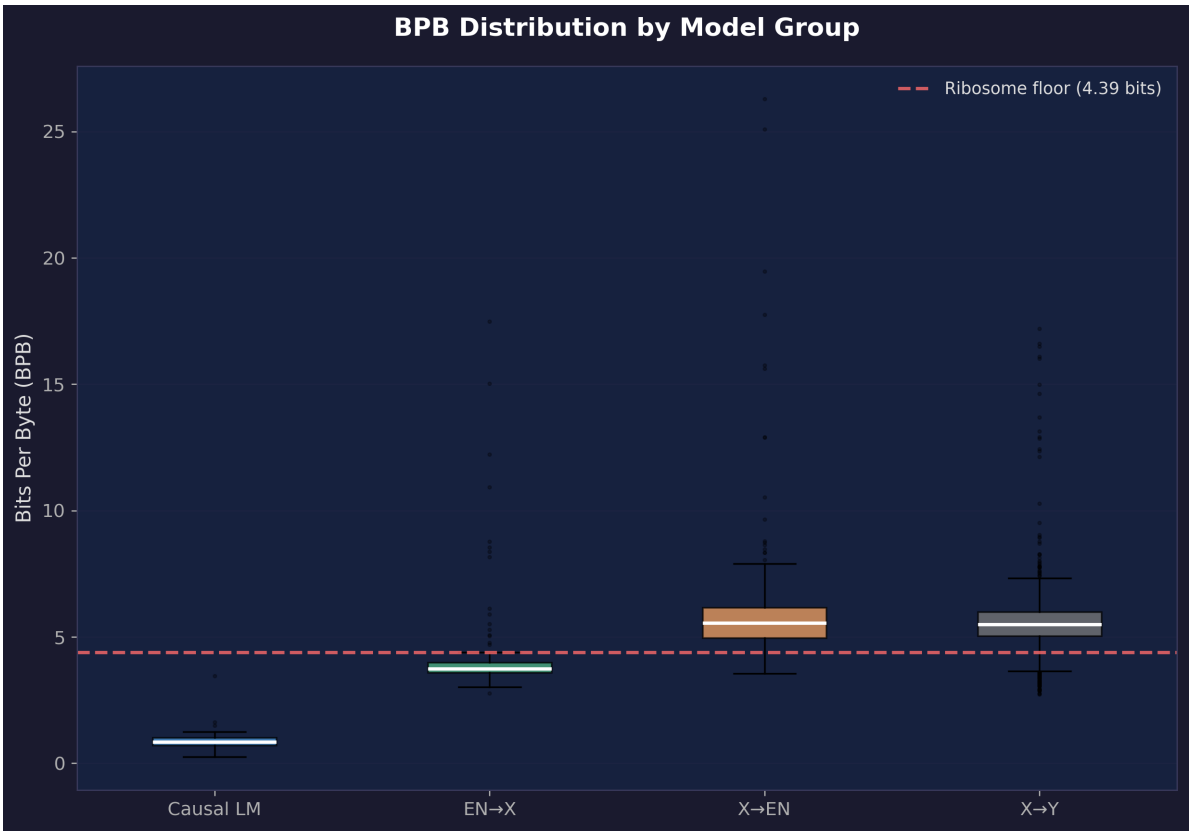

Figure 4: Figure 4: BPB distribution by group — boxplot showing four distinct operating regimes.

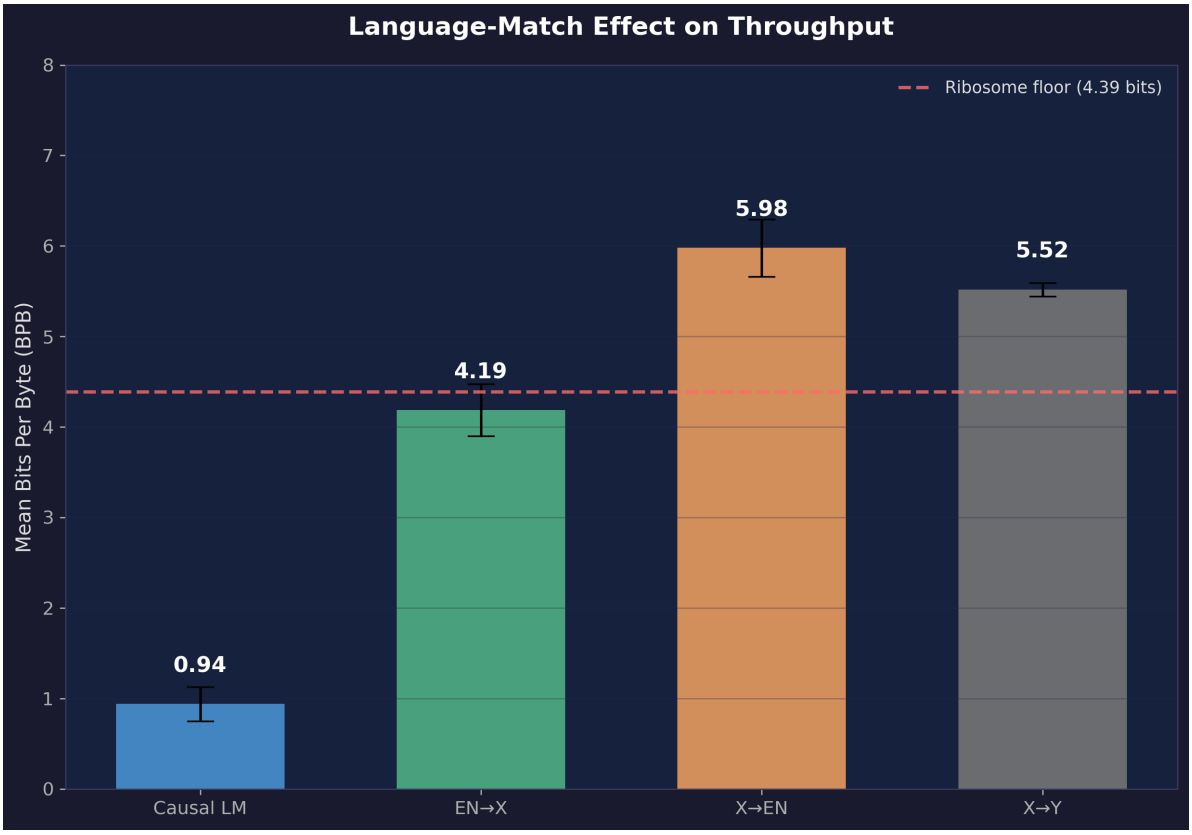

Figure 5: Figure 5: Language-match effect on throughput — mean BPB with 95% CI. EN→X lands at 4.19 bits, near the ribosome floor.

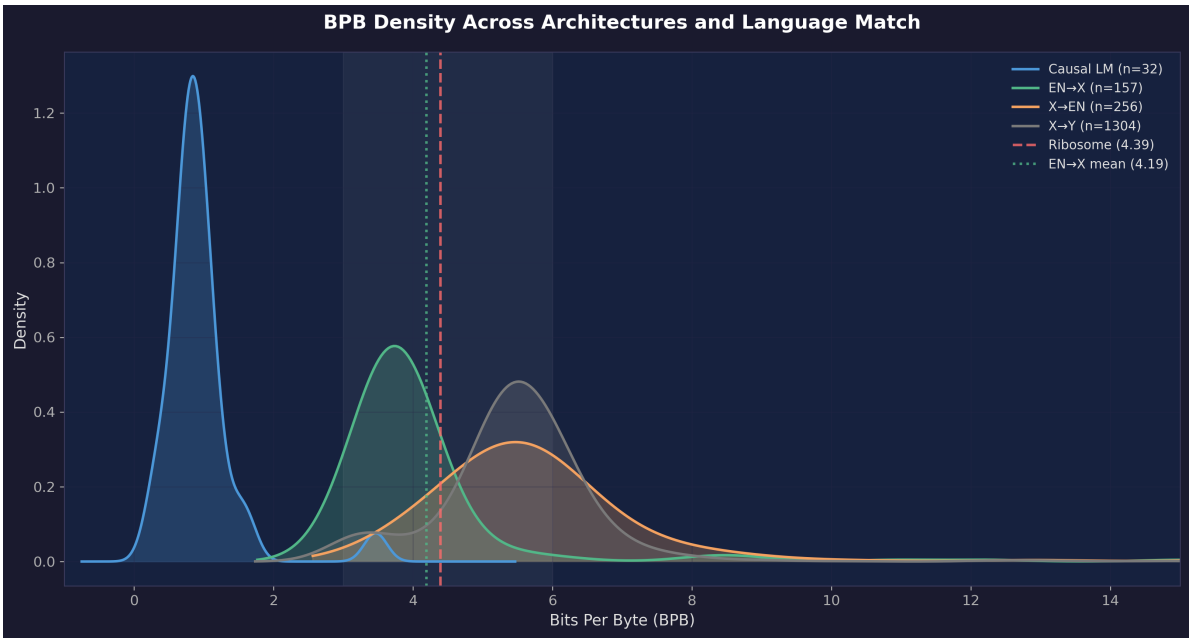

Figure 6: Figure 6: BPB density by group — KDE showing how each group relates to the throughput basin.

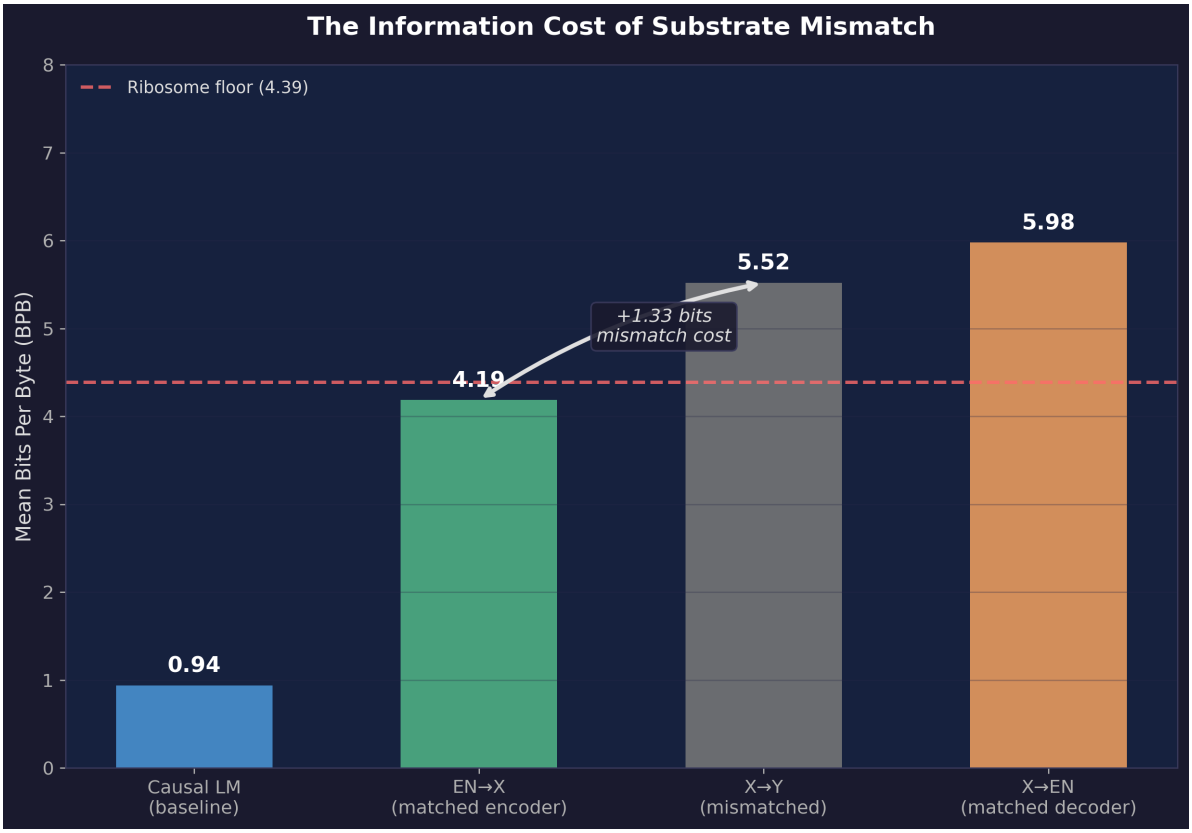

Figure 7: Figure 7: The information cost of substrate mismatch — +1.33 bits between matched and mismatched processing.

3.4 Cross-Architecture Consistency

The vocabulary-independence result holds across both architectures and all four language-match conditions. This is stronger evidence than a single-architecture sweep because it demonstrates that the pattern is not an artifact of any particular model design:

- Causal LMs and translation models operate at different absolute BPB levels (0.94 vs 4.19–5.98)
- Within each level, vocabulary size is not a meaningful predictor
- The receiver-limited hypothesis predicts exactly this: absolute throughput depends on the receiver's capacity and task, but vocabulary is a redundancy parameter in all cases

Vocabulary Independence by Group

Group	n	Statistic	p-value	R^2	Vocab-Independent?
Causal LM	32	β_1=0.058	0.6429	0.0073	YES
EN→X	157	rho=-0.122	0.1274	0.0019	YES
X→EN	256	rho=-0.068	0.2797	0.0158	YES
X→Y	1304	rho=-0.067	0.0148	0.0319	WEAK

Figure 8: Figure 8: Vocabulary independence statistics by group.

3.5 Interpretation

The result is consistent with the receiver-limited hypothesis: vocabulary growth reallocates redundancy and sequence length without increasing per-event throughput. The cross-architecture consistency (causal + encoder-decoder) and the language-match stratification strengthen the claim beyond what a single-architecture sweep could provide.

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4. Empirical Phonology Values

A note on phonology measurements. Two values appear in this trilogy:

- **Support-size bound [SS]:** $\log_2(31) = 4.95$ bits. This is the maximum entropy of the phoneme inventory assuming uniform distribution.
- **Confusion-matrix MI [MI]:** Miller & Nicely (1955) measured $I(X;Y) \approx 4.20$ bits for English consonants — a direct Tier 1 estimate of receiver-resolved information under noise.

The rate-distortion prediction $R_{31}(0.02) = 4.71$ bits sits between these two values. Measured against the support-size bound, $\Delta = +0.24$ bits (the floor predicts lower than the ceiling, as expected). Measured against the MI value, $\Delta = -0.51$ bits (the floor *over*predicts, suggesting that contextual redundancy in real speech reduces effective throughput below even the rate-distortion floor). This discrepancy highlights the importance of the evidence tier system: Tier 1 and Tier 2 measurements of the "same" system can disagree by ~0.75 bits.

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5. Falsifiable Predictions

Prediction 1: Same architecture trained across $V \in \{256, 1K, 4K, 16K, 64K, 256K\}$ — BPB plateaus.

Prediction 2: Tokenizers optimized under corruption converge on smaller, more redundant vocabularies at similar BPB.

Prediction 3: All natural genetic codes encode $15 \leq N \leq 25$ amino acids (Tlusty topological bound).

Prediction 4: Evolutionary simulations converge to 16–32 functional outputs (4–5 bits). Preliminary: three independent implementations yielded $K \approx 19$–30.

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6. Limitations

1. **Near-tautology at low ε.** The ribosome prediction ($\Delta = 0.0004$ under uniform) is near-tautological because $R_M(\varepsilon) \to \log_2 M$ as $\varepsilon \to 0$. Under non-uniform amino acid frequencies, the prediction becomes non-trivial ($\Delta \approx 0.2$). Future work should compute non-uniform R_M for all systems.
2. **Equiprobable symbol assumption.** All predictions assume symmetric error channels with uniform input. Real systems violate both. The robustness analysis (§2.3) suggests the framework survives this violation for the ribosome, but systematic analysis across substrates is needed.
3. **Observational sweep.** The 43-model experiment is observational, not controlled. Architecture, training data, and training duration confound vocabulary-size effects. A controlled single-architecture sweep is the definitive next step.
4. **Tier mixing.** Phonology and music predictions compare $R_M(\varepsilon)$ against support-size bounds [SS], not mutual information [MI]. Miller & Nicely (1955) confusion matrices give $I(X;Y) \approx 4.22$ bits for English consonants — a Tier 1 measurement. Conditional entropy rates from Conklin (1995), Temperley (2014), and Pimentel (2020) show that music operates at 1.87–2.97 bits/note and phonemes at ~3.0 bits — significantly lower than support-size values. If the analysis is redone with conditional entropy, the convergence band may shift downward to approximately [1.5, 4.5] bits. This does not invalidate the rate-distortion framework but changes its empirical calibration.

5. **AI I_eff is model-dependent**, ranging from ~1.4 (frontier) to ~5+ (small models). There is no single substrate-level "AI throughput."

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7. Conclusion

The rate-distortion floor R_M(ε) provides a universal mechanistic bound for serial decoding throughput. For biological receivers with moderate M and moderate ε, this bound confines throughput to 3–5 bits — a geometric property of the rate-distortion surface, not an empirical coincidence. The empirical finding that AI bits-per-byte is independent of vocabulary size across 1,749 models (causal LMs: p = 0.643; matched-encoder translation: Spearman p = 0.13) supports the vocabulary-as-redundancy hypothesis. The rate-distortion framework, the biological predictions, and the AI regularity constitute three independent lines of evidence for a receiver-limited serial decoding constraint.

References

Berger, T. (1971). *Rate Distortion Theory.* Prentice-Hall.

Borst, A., & Theunissen, F. E. (1999). Information theory and neural coding. *Nature Neuroscience*, 2(11), 947–957.

Hopfield, J. J. (1974). Kinetic proofreading. *PNAS*, 71(10), 4135–4139.

Miller, G. A., & Nicely, P. E. (1955). Perceptual confusions among English consonants. *JASA*, 27(2), 338–352.

Shannon, C. E. (1948). A mathematical theory of communication. *Bell Sys. Tech. J.*, 27(3), 379–423.

Tlusty, T. (2008). Rate-distortion scenario for noisy molecular codes. *Phys. Rev. Lett.*, 100, 048101.

Whitmer III, G.L. (2026a). The Fons Constraint. Zenodo. doi:10.5281/zenodo.19274048.

Zaher, H. S., & Green, R. (2009). Quality control by the ribosome. *Nature*, 457(7226), 161–166.

Acknowledgments

This paper was improved through adversarial review by six frontier AI models. The tokenizer-sweep experiment was executed by Hermes OC1 on an NVIDIA RTX 5090.

Appendix A: Experiment Protocol

The complete, verbatim instructions given to the experiment operator (Hermes OC1) are provided in the supplementary file `APPENDIX-A-EXPERIMENT-PROTOCOL.md`. This in-

cludes hardware specifications, corpus preparation, BPB computation methods for three model formats (HuggingFace, GGUF, MarianMT), statistical analysis procedures, and visualization specifications. The protocol was designed for full reproducibility: any researcher with access to HuggingFace models and a CUDA-capable GPU can replicate the experiment using the provided instructions.

Data Availability

https://github.com/sneakyfree/fons-constraint

Raw experiment data: `results/bpb_results.csv`, `results/model_catalog.csv`, `results/statistical_summary.txt`

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